

AI Internship – Day 1 Report

Exploratory Data Analysis (EDA) on Titanic Dataset

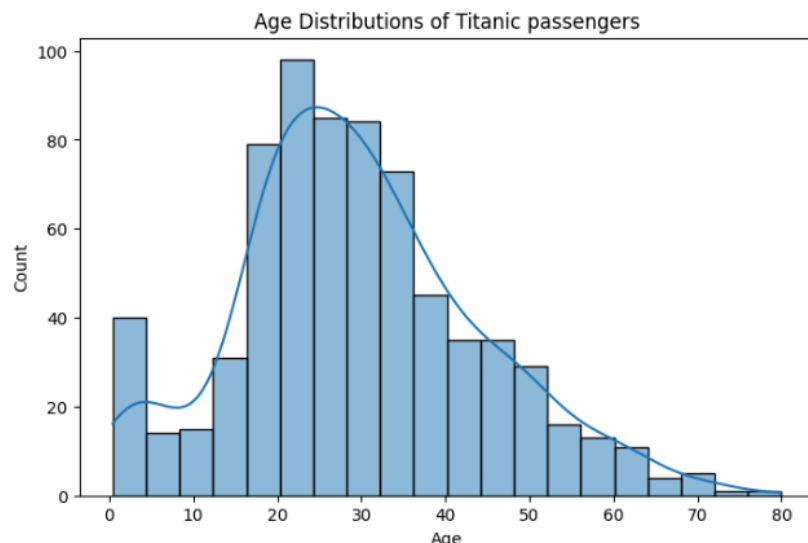
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Date: 2/6/2026

Objective:

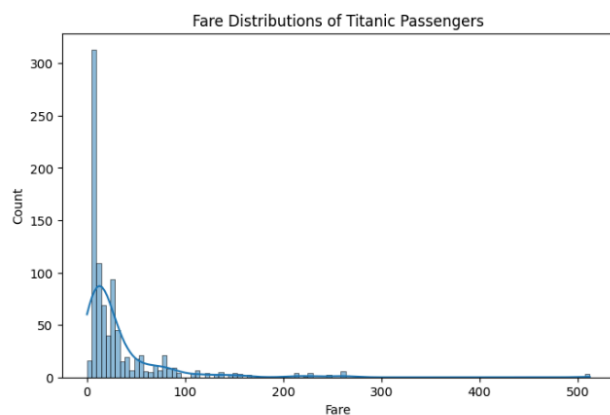
To set up a machine learning environment and perform exploratory data analysis (EDA) on the Titanic dataset using Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn.

Findings

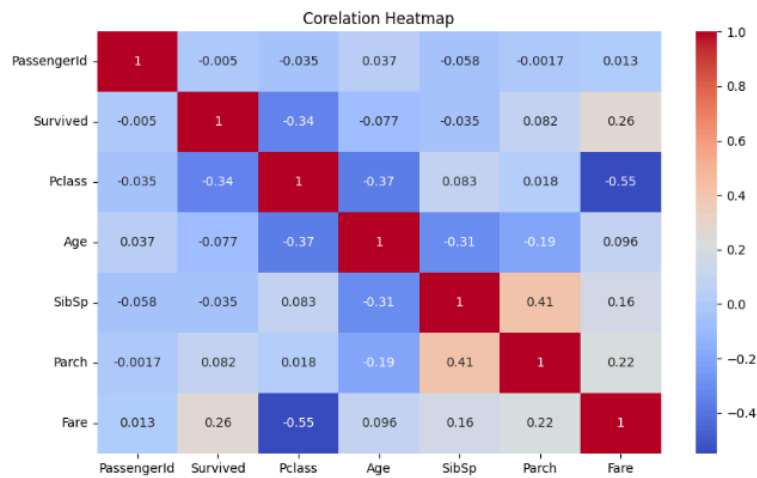
1. The Titanic dataset contains 891 rows and 12 features including passenger details and survival information.
2. Most passengers are between 20–40 years old based on the age distribution plot.



3. Fare distribution is highly right-skewed, indicating most passengers bought low-cost tickets while few paid very high fares.



4. Correlation analysis shows that Fare positively influences survival, while Pclass negatively affects survival chances.



5. Missing Values Heatmap shows Age has 177 missing values and Cabin has a large number of missing values (687), making it a low-quality feature.

- Cabin column shows maximum missing data

- Age shows scattered missing values

